MATH0057 Probability and Statistics

Year:	2026–2027
Code:	MATH0057
Level:	5 (UG)
Normal student group(s):	UG: Y2 Mathematics programmes
Value:	15 credits (7.5 ECTS credits)
Term:	2
Assessment:	90% examination, 10% coursework
Normal Pre-requisites:	MATH0011 (Mathematical Methods 2)
Lecturer:	Dr I Strouthos

Course Description and Objectives

The aim of the course is to introduce students to the theory of probability and some of the statistical methods based upon it. Many physical processes involve random components which can only be modelled using probabilistic methods. Statistical theory is vital for analysing scientific data where it is necessary to distinguish genuine patterns from random fluctuations.

Recommended Texts

- Wackerley, Mendenhall and Scheaffer, *Mathematical Statistics with Applications* 6th ed (Duxbury)

Detailed Syllabus

- Axiomatic approach to probability; standard discrete and continuous distributions (Bernoulli, binomial, geometric, negative binomial, hypergeometric, Poisson, uniform, exponential, gamma, beta, normal), their properties and uses; idea of Poisson process.
- Joint probability distributions: joint and conditional distributions and moments; iterated expectation; multinomial and multivariate normal distributions.
- Moment and probability generating functions; properties; sums of independent random variables.
- Statement of weak law of large numbers.
- Statement of the Central Limit Theorem.
- Introduction to statistics. Data, estimation and hypothesis testing.
- Normal probability models. χ^2, t and F distributions. Confidence intervals.
- Regression and correlation. Least-squares.

September 2026